Q.1 A loop at a state is supposed to be ---------- transition while converting Moore machine into an equivalent Melay machine.

A. Both incoming and outgoing

B. Incoming

C. Outgoing

D. Complex

Q.2 A non regular language can be represented by

A. TG

B. RE

C. FA

D. None of the given options

Q.3 The complement of a regular language is also _____.

A. regular

B. irregular

C. irregular but finite

D. irregular but infinite

Q.4 Which one of the following machine is represented as a pictorial representation with states and directed edges labeled by an input letter along with an output character?

A. Finite state machine

B. Mealy machine

C. Moore machine

D. Deterministic finite state machine

Q.5 If R is regular language and Q is any language (regular/ non-regular), then Pref(_____in _____) is regular.

A. R,R

B. Q,R

C. Q,Q

D. R,Q

Q.6 If an FA accepts a word then there must exist a path from _____.

A. Initial to final state by traversing each state

B. Initial to final state

C. Initial to each state

D. Initial to each state but not to final state

Q.7 Finite Automaton (FA) must have _____ number of states while a language has _____ words.

A. finite, finite

B. infinite, finite

C. infinite, infinite

D. finite, infinite

Q.8 The CFG is said to be ambiguous if there exist at least one word of its language that can be generated by _____ production tree(s).

A. More than one

B. At most one

C. One

D. Zero

Q.9 The language L defined over Σ, not belonging to L, is called _____ of the language L.

A. complement

B. union

C. intersection

D. closure

Q.10 Which of the following represent the absence and presence of current in sequential circuit respectively?

A. 0, 1

B. 1, 1

C. 1, 0

D. 0, 0
